Pinching a free normal cover: variable critical exponent in reduced quasiconformal Teichmüller space

Anonymous candidate contribution

1 September 2026

Abstract

We give an explicit all-Fuchsian strengthening candidate related to M. Kapovich's AIM Problem 4.3. For every closed oriented surface S_g of genus $g \geq 2$, fix the kernel K of the epimorphism $\pi_1(S_g) \to F_g$ that kills a disjoint cut system. For every marked closed hyperbolic metric m, the image Γ_m of K under the Fuchsian holonomy is an infinitely generated Fuchsian group of the first kind. Marked quasiconformal maps of the compact base lift to quasiconformal conjugacies of the groups Γ_m, so this gives a family in one reduced, surface-based quasiconformal Teichmüller space. The quotient F_g is nonamenable; the normal-subgroup theorem of Dougall and Sharp therefore gives $\delta(\Gamma_m) < 1$ for every marked compact-base metric. If the cut curves are simultaneously pinched to common length ℓ, a compactly supported function on one lifted cut-surface cell has Rayleigh quotient $O_g(\ell)$. The Elstrodt–Patterson–Sullivan formula then yields

$$\frac{1}{2} < \delta(\Gamma_{m_\ell}) < 1, \qquad 1 - \delta(\Gamma_{m_\ell}) = O_g(\ell), \qquad \delta(\Gamma_{m_\ell}) \longrightarrow 1.$$

Thus the critical exponent assumes infinitely many distinct values among the all-Fuchsian points of $T_{\mathrm{qc}}^{\mathrm{red}}(X_0)$. Under the reduced surface-based reading of the AIM question this is a direct solution candidate; under the broader group-deformation reading it is an all-Fuchsian strengthening of an affirmative precedent of Astala and Zinsmeister.

中文摘要（繁體）

本文提出 M. Kapovich 之 AIM 問題 4.3 的一個明確解答候選。對每個虧格 $g \geq 2$ 的閉可定向曲面 S_g，固定滿同態 $\pi_1(S_g) \to F_g$ 的核 K；此滿同態消去一組互不相交的割曲線。對任一有標記的閉雙曲度量 m，K 在 Fuchsian 單值群中的像 Γ_m 是第一類且無限生成的 Fuchsian 群。緊緻底曲面之間的有標記 擬共形映射可提升為 Γ_m 之間的擬共形共軛，故所有這些群位於同一 約化擬共形 Teichmüller 空間。商群 F_g 非可均，因此 Dougall--Sharp 定理給出每一 有限階段的 $\delta(\Gamma_m) < 1$。若同時把割曲線壓縮至共同長度 ℓ， 則一個提升之割開胞腔上的緊支撐函數具有 $O_g(\ell)$ 的 Rayleigh 商。Elstrodt--Patterson--Sullivan 公式遂推出 $1/2 < \delta(\Gamma_{m_\ell}) < 1$，且 $\delta(\Gamma_{m_\ell}) \to 1$。因此臨界指數在同一擬共形變形空間中取得 無限多個不同值。

Translation note. The Traditional Chinese abstract is a producer-generated rendering and was not independently assessed by a Chinese language reviewer; the English abstract controls if the two differ.

Status. This is an anonymous, AI-assisted, unrefereed all-Fuchsian strengthening candidate. It includes a complete written proof and producer-side formula checks, but no claim of historical priority, external specialist validation, independent reconstruction, formal proof verification, or peer review. Candidate publication and DOI assignment do not alter those assurance limits.

2020 Mathematics Subject Classification. Primary 30F35; Secondary 30F60, 58J50. **Keywords.** Fuchsian group, critical exponent, quasiconformal Teichmüller space, normal cover, bottom of spectrum, collar degeneration.

Contents

1 Introduction

Let Γ_0 be an infinitely generated Fuchsian group of the first kind. AIM Problem 4.3, attributed to M. Kapovich, states:

> "Construct an infinitely generated Fuchsian group Γ_0 of the first kind such that the critical exponents of some elements in $T_{qc}(\Gamma_0)$ are different."

The problem arose at the 2012 AIM workshop, appears in the accessible AIM problem document dated 2014, and was published in handbook form in 2016 (Su, 2014, 2016). This question is meaningful because first kind means that the limit set is the entire boundary circle, whereas for an infinitely generated group the critical exponent can nevertheless be strictly smaller than one. The critical exponent measures exponential orbit growth: roughly, how rapidly the number of group translates within distance R grows with R. A first-kind group has the full boundary circle as limit set, but for an infinitely generated group this topological fact does not force exponent one. Pinching the cut curves creates long collars, so a function concentrated on one lifted cell has very low spectral energy; the spectral formula then drives the exponent toward one. Examples with exponent below one occur elsewhere (Bishop, 2003; Lehnert, 2017).

There is an important scope boundary. Under a broad group-deformation reading, the quasi-Fuchsian family of Astala and Zinsmeister (1995) appears already to give an affirmative existential example. We therefore make no first-solution or priority claim. The contribution here is a fixed-kernel construction that stays entirely in the all-Fuchsian locus, has every non-degenerate exponent below one, and approaches one with an explicit one-sided rate. Under the narrower reduced surface-based, all-Fuchsian reading of the AIM question, it is a direct solution candidate.

The construction here uses a fixed normal cover of a closed surface. Its deck group is a nonabelian free group. Two competing consequences of that choice drive the proof. At every fixed metric, nonamenability makes the normal subgroup's critical exponent strictly smaller than the ambient lattice exponent. On the other hand, when a complete cut system is pinched, one fundamental cell becomes separated from its neighbours by increasingly long collars. A function supported on that cell has arbitrarily small Dirichlet energy. The spectral formula converts this geometric degeneration into critical exponents tending to one.

We use the nonnegative Laplacian. For a complete surface X, its spectral bottom is

$$\lambda_0(X) = \inf_{0\neq f\in C_c^\infty(X)} \frac{\int_X |\nabla f|^2\, dA}{\int_X |f|^2\, dA}.$$

For a nonelementary Fuchsian group Γ, the critical exponent is

$$\delta(\Gamma) = \inf\left\{ s>0 : \sum_{\gamma\in\Gamma} e^{-sd(o,\gamma o)} < \infty \right\},$$

independent of $o \in \mathbb{H}^2$.

Fix a base metric m_0 and put $X_0 = X_{m_0}$. We use the reduced, surface-based convention of Alessandrini et al. (2012): a point of $T_{\mathrm{qc}}^{\mathrm{red}}(X_0)$ is an equivalence class of a marked conformal surface (Y, f), where $f : X_0 \to Y$ is quasiconformal, and (Y_1, f_1) and (Y_2, f_2) are equivalent when a conformal map $c : Y_1 \to Y_2$ is homotopic to $f_2 \circ f_1^{-1}$. In the reduced convention boundary components are preserved setwise rather than pointwise; the surfaces here have no boundary, so that distinction is moot. The theorem concerns the Fuchsian uniformising groups of the marked surfaces constructed below. It does not quantify over arbitrary quasi-Fuchsian sphere conjugates.

For comparison with the original notation, write $T_{\mathrm{qc}}(\Gamma_0)$ for the reduced group quasiconformal deformation space: a quasiconformal sphere homeomorphism W conjugating Γ_0 to a Kleinian group determines a point, and marked representatives that differ by Möbius conjugacy determine the same point. This is the only group-space structure used below. Möbius conjugacy preserves the critical exponent, since it acts isometrically on hyperbolic three-space. Lemma 2 constructs such a sphere representative for every marked compact-base metric and shows that the entire family lies in the all-Fuchsian locus of $T_{\mathrm{qc}}(\Gamma_0)$. Thus the construction produces points for the AIM notation without asserting a global identification of every group deformation with the surface-based space.

2 The fixed free normal cover

Fix $g \geq 2$. Choose a geometric symplectic basis $(a_1, b_1, \dots, a_g, b_g)$ for S_g so that the simple curves $b_1, \dots, b_g$ are pairwise disjoint and their complement is connected. Thus

$$\pi_1(S_g) = \left\langle a_1, b_1, \dots, a_g, b_g \;\middle|\; \prod_{i=1}^{g} [a_i, b_i] = 1 \right\rangle.$$

Let $F_g = \langle x_1, \dots, x_g \rangle$ and define

$$q : \pi_1(S_g) \longrightarrow F_g, \qquad q(a_i) = x_i, \qquad q(b_i) = 1. \tag{1}$$

The surface relator maps to the identity, and the images of the a_i generate F_g; hence q is an epimorphism. Put $K = \ker q$. Thus there is an exact sequence

$$1 \longrightarrow K \longrightarrow \pi_1(S_g) \xrightarrow{q} F_g \longrightarrow 1.$$

For a marked closed hyperbolic metric m on S_g, choose its faithful Fuchsian holonomy

$$\rho_m : \pi_1(S_g) \longrightarrow \mathrm{PSL}(2, \mathbb{R}),$$

write $L_m = \rho_m(\pi_1(S_g))$, and set

$$\Gamma_m = \rho_m(K), \qquad X_m = \mathbb{H}^2/\Gamma_m.$$

Then $X_m \to S_m = \mathbb{H}^2/L_m$ is the regular cover with deck group F_g. Equivalently,

$$1 \longrightarrow \Gamma_m \longrightarrow L_m \longrightarrow F_g \longrightarrow 1.$$

Lemma 1. *For every m, the group Γ_m is nonelementary, infinitely generated, and of the first kind.*

Proof. The subgroup K is nontrivial because it contains each b_i. Hence Γ_m contains a hyperbolic element. Its limit set is nonempty and, because $\Gamma_m \lhd L_m$, is invariant under L_m. The limit set of the cocompact lattice L_m is the full circle and is the minimal nonempty closed L_m-invariant subset of the boundary. It follows that

$$\Lambda(\Gamma_m) = \Lambda(L_m) = \partial\mathbb{H}^2.$$

In particular Γ_m is nonelementary and of the first kind.

Suppose that Γ_m were finitely generated. A finitely generated Fuchsian group without parabolics is convex cocompact (see, for example, Beardon, 1983). There are no parabolics here because Γ_m is a subgroup of the cocompact torsion-free group L_m. Since its limit set is the full circle, its convex hull is all of $\mathbb{H}^2$; convex cocompactness would therefore make $X_m = \mathbb{H}^2/\Gamma_m$ compact. A compact cover of the compact surface S_m has finite degree. This contradicts $L_m/\Gamma_m \cong F_g$, which is infinite. Thus Γ_m is infinitely generated. □

Lemma 2. *Fix a marked metric m_0. For every marked closed hyperbolic metric m, the lifted marked surface X_m represents a point of $T_{\mathrm{qc}}^{\mathrm{red}}(X_0)$, and its Fuchsian uniformising group is Γ_m. Consequently the family $\{\Gamma_m : m \in \mathrm{Teich}(S_g)\}$ is a family of Fuchsian points in one reduced quasiconformal Teichmüller space.*

Proof. There is an orientation- and marking-preserving quasiconformal map $h : S_{m_0} \to S_m$. Choose the lift $\tilde{h} : \mathbb{H}^2 \to \mathbb{H}^2$ that is equivariant for the marked holonomies. Then

$$\tilde{h}\,\rho_{m_0}(\gamma)\,\tilde{h}^{-1} = \rho_m(\gamma) \qquad (\gamma \in \pi_1(S_g)).$$

Restricting this identity to the fixed subgroup K gives $\tilde{h}\Gamma_{m_0}\tilde{h}^{-1} = \Gamma_m$. The lift is quasiconformal and descends to the marked map $X_0 \to X_m$, so it gives the asserted point of $T_{\mathrm{qc}}^{\mathrm{red}}(X_0)$. The markings identify the same abstract kernel $K = \ker q$ at both ends. Its boundary extension is quasisymmetric, and reflection across the boundary circle supplies a quasiconformal sphere extension W_m satisfying $W_m\Gamma_{m_0}W_m^{-1} = \Gamma_m$. Hence it also gives a point of $T_{\mathrm{qc}}(\Gamma_{m_0})$ in the original AIM notation. If two surface representatives are equivalent, the conformal equivalence lifts to a Möbius conjugacy of their uniformising groups; consequently their critical exponents agree. That observation does not enlarge the theorem to arbitrary quasi-Fuchsian conjugates. No injectivity claim for the induced map of Teichmüller spaces is needed. □

3 Strict comparison with the ambient lattice

The ambient group L_m is cocompact, hence convex cocompact, and $\delta(L_m) = 1$. The quotient

$$L_m/\Gamma_m \cong F_g$$

is nonamenable because $g \geq 2$. The normal-subgroup theorem of Dougall and Sharp (2016, Theorem 1.1) says that for a normal subgroup of a convex-cocompact group acting on a pinched Hadamard manifold, equality of critical exponents holds exactly when the quotient is amenable. Here $\mathbb{H}^2$ is complete, simply connected, and has curvature -1; L_m is nonelementary and cocompact, hence convex cocompact; $\Gamma_m \lhd L_m$; and $L_m/\Gamma_m \cong F_g$ is nonamenable. Every hypothesis is therefore discharged, and

$$\delta(\Gamma_m) < \delta(L_m) = 1 \qquad \text{for every marked metric } m. \tag{2}$$

This is the only place where nonamenability is used. Equation (2) also chooses the correct branch when the spectral formula is applied after degeneration.

4 Pinching the cut system

Extend $b_1, \dots, b_g$ to a pants decomposition. Fenchel–Nielsen coordinates give a path of marked closed hyperbolic metrics m_ℓ, $0 < \ell < \ell_*$, for which

$$\ell_{m_\ell}(b_i) = \ell \qquad (1 \le i \le g).$$

The remaining length and twist coordinates may be held fixed (Buser, 1992). Let $X_\ell = X_{m_\ell}$ and $\Gamma_\ell = \Gamma_{m_\ell}$.

Cutting S_g along the b_i produces a sphere P with $2g$ boundary components. The cover defined by (1) can be constructed from copies of P indexed by F_g: after orienting the two sides of each b_i, the corresponding sides of the copies labelled h and hx_i are glued. Indeed, the curve a_i is dual to b_i: after the cut it becomes an arc joining the two boundary copies of b_i, crosses no other cut curve, and its monodromy is $q(a_i) = x_i$. In particular, X_ℓ contains an embedded compact lifted cell P_ℓ.

The collar lemma supplies pairwise disjoint open collars around the b_i with half-width

$$w = w(\ell) = \operatorname{arcsinh}\left(\frac{1}{\sinh(\ell/2)}\right) \longrightarrow \infty \qquad (\ell \downarrow 0) \tag{3}$$

and metric

$$ds^2 = dr^2 + \ell^2 \cosh^2 r \, d\theta^2, \qquad -w < r < w, \quad \theta \in \mathbb{R}/\mathbb{Z} \tag{4}$$

(Keen, 1974; Buser, 1992). Each cut cell contains $2g$ half-collars, parametrized by $0 \le r \le w$, with the cut geodesic at $r = 0$.

For sufficiently small ℓ, $w(\ell) > 1$. Define $u_\ell : X_\ell \to [0,1]$ as follows. On each of the $2g$ selected-cell half-collars put

$$u_\ell(r,\theta) = \begin{cases} r, & 0 \le r \le 1, \\ 1, & 1 \le r < w(\ell). \end{cases}$$

Put $u_\ell = 1$ on the remainder of that cell and $u_\ell = 0$ outside it. The traces agree across every seam because $u_\ell = 0$ at $r = 0$. Thus u_ℓ is a compactly supported Lipschitz function. It belongs to $H^1_c(X_\ell)$ and can be approximated in H^1 by functions in $C_c^\infty(X_\ell)$, so it is admissible in the Rayleigh principle.

Proposition 3. *There are constants $C_g > 0$ and $\ell_g > 0$ such that*

$$0 \le \lambda_0(X_\ell) \le C_g \ell \qquad (0 < \ell < \ell_g).$$

In particular $\lambda_0(X_\ell) \to 0$.

Proof. The gradient is supported in $0 < r < 1$, where $|\nabla u_\ell|^2 = 1$. The area element from (4) is $\ell \cosh r\, dr\, d\theta$. One half-collar therefore contributes

$$\int_0^1 \int_0^1 \ell \cosh r\, dr\, d\theta = \ell \sinh 1.$$

Summing over $2g$ half-collars gives the exact numerator

$$E_\ell = \int_{X_\ell} |\nabla u_\ell|^2\, dA = 2g\ell \sinh 1. \tag{5}$$

By Gauss–Bonnet, $\operatorname{area}(S_{m_\ell}) = 4\pi(g-1)$. Each of the $2g$ width-one transition strips has area $\ell \sinh 1$. Hence the plateau on which $u_\ell = 1$ has area at least

$$D_\ell = 4\pi(g-1) - 2g\ell \sinh 1. \tag{6}$$

For all sufficiently small ℓ, specifically whenever $2g\ell \sinh 1 \leq 2\pi(g-1)$, we have $D_\ell \geq 2\pi(g-1) > 0$, and therefore

$$\int_{X_\ell} u_\ell^2\, dA \geq D_\ell \geq 2\pi(g-1).$$

The Rayleigh principle and (5) give

$$\lambda_0(X_\ell) \leq \frac{E_\ell}{\int u_\ell^2} \leq \frac{g \sinh 1}{\pi(g-1)}\, \ell.$$

Thus one may take $C_g = g \sinh 1/[\pi(g-1)]$ after decreasing ℓ_g if necessary. □

5 Critical exponents and the all-Fuchsian conclusion

For any nonelementary Fuchsian group Γ, the Elstrodt–Patterson–Sullivan formula, in the nonnegative-Laplacian convention, is

$$\lambda_0(\mathbb{H}^2/\Gamma) = \begin{cases} \frac{1}{4}, & \delta(\Gamma) \leq \frac{1}{2}, \\ \delta(\Gamma)(1-\delta(\Gamma)), & \delta(\Gamma) \geq \frac{1}{2}. \end{cases} \tag{7}$$

See Sullivan (1987, Theorem 2.17, p. 333). Its assumptions are met: Γ_ℓ is torsion-free, discrete, and nonelementary; X_ℓ is a complete curvature--1 surface; and our λ_0 uses the nonnegative Laplacian. We can now state the result.

Theorem 4 (All-Fuchsian strengthening candidate for AIM Problem 4.3)**.** *For every $g \geq 2$, the group $\Gamma_0 = \Gamma_{m_0}$ constructed above is an infinitely generated Fuchsian group of the first kind for which the critical exponent takes infinitely many distinct values among the Fuchsian points of $T_{\mathrm{qc}}^{\mathrm{red}}(X_0)$. More precisely, along the simultaneous pinching family, for all sufficiently small ℓ,*

$$\frac{1}{2} < \delta(\Gamma_\ell) < 1, \qquad 0 < 1 - \delta(\Gamma_\ell) \leq 2C_g\ell,$$

and $\delta(\Gamma_\ell) \to 1$ as $\ell \downarrow 0$.

Proof. Lemmas 1 and 2 give the required group and place the whole family in $T_{\mathrm{qc}}^{\mathrm{red}}(X_0)$. By Proposition 3, $\lambda_0(X_\ell) < 1/4$ for all sufficiently small ℓ. Formula (7) then rules out $\delta(\Gamma_\ell) \leq 1/2$, so

$$\lambda_0(X_\ell) = \delta(\Gamma_\ell)(1 - \delta(\Gamma_\ell)).$$

Equation (2) supplies $\delta(\Gamma_\ell) < 1$. Consequently

$$0 < 1 - \delta(\Gamma_\ell) = \frac{\lambda_0(X_\ell)}{\delta(\Gamma_\ell)} < 2\lambda_0(X_\ell) \le 2C_g \ell.$$

This proves $\delta(\Gamma_\ell) \to 1$ from below. No exponent belonging to a marked compact-base metric is one, so the values along any sequence $\ell_n \downarrow 0$ cannot form a finite set: a finite set contained in $(-\infty, 1)$ has maximum strictly smaller than one. Thus infinitely many values are distinct, and in particular the critical exponent is nonconstant among the Fuchsian points of $T_{\mathrm{qc}}^{\mathrm{red}}(X_0)$. □

Remark 5. The proof does not show that $\delta(\Gamma_\ell)$ is monotone in ℓ. It also does not identify an exact formula for the exponent. The explicit rate is one-sided and is inherited from a single fixed-width test function. The symbol $\ell = 0$ denotes a boundary degeneration, not a metric or a point of the asserted family.

6 Relation to previous work

Astala and Zinsmeister (1995) used a normal cover of a compact genus-three surface with abelian deck group $\mathbb{Z}^3$ and holomorphic quasi-Fuchsian deformations to exhibit variation of the Poincaré exponent, including values above one. Under the broad group-deformation interpretation of $T_{\mathrm{qc}}(\Gamma_0)$, this appears already to answer the existential AIM question affirmatively. The present theorem is therefore positioned as an all-Fuchsian strengthening and alternative construction, and as a direct solution candidate only under the reduced surface-based, all-Fuchsian reading. It is not the same family: here the deck group is the nonamenable free group F_g, every compact-base exponent is below one, and pinching makes the exponents approach one from below. Bishop's δ-stability results and the later Ruelle-property work concern the relation between exponent, limit-set dimension, and quasi-Fuchsian deformation (Bishop, 2003; Huo and Zinsmeister, 2022). Regular covers and the sharp lower threshold 1/2 for their exponents are studied by Bonfert-Taylor et al. (2012). These results delimit the contribution: the claim here is the fixed-kernel, all-Fuchsian pinching construction in reduced surface-based quasiconformal Teichmüller space, not a general deformation or stability theorem.

7 Dependency and limitation statement

The family construction, quasiconformal lift, collar function, energy integral, area estimate, and final algebra are given explicitly. The proof imports two decisive structural results: the Dougall–Sharp normal-subgroup amenability criterion and the Elstrodt–Patterson–Sullivan spectral formula. Dougall–Sharp Theorem 1.1 is applied with $X = \mathbb{H}^2$, $\Gamma = L_m$, and $\Gamma_0 = \Gamma_m$; Sullivan Theorem 2.17 is applied in dimension $d = 1$ to $\mathbb{H}^2/\Gamma_\ell$. The hypothesis maps are also recorded in `PROOF_OBLIGATIONS.md`. The argument uses standard finite-type Teichmüller and collar theorems. The accompanying program recomputes only the explicit collar formulas for sample parameters; it does not verify these external theorems or replace the universal written argument.

A bounded search through the AIM source, arXiv, publisher and author-hosted primary papers located the adjacent work just described but did not locate this exact fixed free-normal-cover, all-Fuchsian pinching statement. That negative search evidence is not a priority claim. Specialist review and exhaustive MathSciNet/zbMATH searching remain open assurance steps.

Data and materials availability

There are no empirical data. The local research package contains the LaTeX and Markdown manuscripts, bibliography, source and claim ledgers, standard-library formula checker, mutation tests, build scripts, PDF, and cryptographic manifest. The PDF is visually checked but untagged; the Markdown manuscript is the accessible-text fallback, with the remaining limitation that its mathematics is linearized plain text rather than fully semantic math. Public repository and archival identifiers, when assigned, are recorded in the release metadata; their existence does not increase the mathematical assurance level.

Ethics statement

This theoretical mathematics work uses no human participants, personal data, animals, clinical material, or field interventions. No ethics approval was required.

Author contributions (CRediT)

Anonymous candidate contribution: Conceptualization, Formal analysis, Methodology, Software, Validation, Writing—original draft, and Writing—review & editing. Operational provenance and AI assistance are reported separately below; no AI system is listed as an author.

Funding

No external funding was declared for this candidate.

Competing interests

No competing interests were declared.

AI-use disclosure

OpenAI Codex was used to search and inspect sources, develop and stress-test the construction, draft and typeset the manuscript, write corroborative checks, and assemble the local package. The workflow is producer-side and does not constitute independent mathematical validation. A human publisher or reviewer should verify the proof, citations, attribution, and novelty boundary before any public release.

References

Daniele Alessandrini, Lixin Liu, Athanase Papadopoulos, and Weixu Su. On various teichmüller spaces of a surface of infinite topological type. *Proceedings of the American Mathematical Society*, 140(2):561–574, 2012. doi: 10.1090/S0002-9939-2011-10918-3. URL `https://arxiv.org/abs/1008.2851`.

Kari Astala and Michel Zinsmeister. Abelian coverings, poincaré exponent of convergence and holomorphic deformations. *Annales Academiae Scientiarum Fennicae. Series A I. Mathematica*, 20:81–86, 1995. URL `https://www.acadsci.fi/mathematica/Vol20/astala.pdf`.

Alan F. Beardon. *The Geometry of Discrete Groups*, volume 91 of *Graduate Texts in Mathematics*. Springer, New York, 1983. doi: 10.1007/978-1-4612-1146-4.

Christopher J. Bishop. δ-stable fuchsian groups. *Annales Academiae Scientiarum Fennicae Mathematica*, 28:153–167, 2003. URL `https://www.acadsci.fi/mathematica/Vol28/bishop.pdf`.

Petra Bonfert-Taylor, Katsuhiko Matsuzaki, and Edward C. Taylor. Large and small covers of a hyperbolic manifold. *Journal of Geometric Analysis*, 22(2):455–470, 2012. doi: 10.1007/s12220-010-9204-6. URL `https://matsuzak.w.waseda.jp/Preprint/large-small.pdf`.

Peter Buser. *Geometry and Spectra of Compact Riemann Surfaces*, volume 106 of *Progress in Mathematics*. Birkhäuser, Boston, 1992. doi: 10.1007/978-0-8176-4992-0.

Rhiannon Dougall and Richard Sharp. Amenability, critical exponents of subgroups and growth of closed geodesics. *Mathematische Annalen*, 365(3–4):1359–1377, 2016. doi: 10.1007/s00208-015-1338-1. URL `https://arxiv.org/abs/1411.6817`.

Shengjin Huo and Michel Zinsmeister. On ruelle's property. *Ergodic Theory and Dynamical Systems*, 42(4):1474–1486, 2022. doi: 10.1017/etds.2020.149. URL `https://doi.org/10.1017/etds.2020.149`.

Linda Keen. Collars on riemann surfaces. In Leon Greenberg, editor, *Discontinuous Groups and Riemann Surfaces*, volume 79 of *Annals of Mathematics Studies*, pages 263–268. Princeton University Press, 1974. doi: 10.1515/9781400881642-021.

Ralf Lehnert. On the critical exponent of infinitely generated veech groups. *Mathematische Annalen*, 368(3–4):1017–1058, 2017. doi: 10.1007/s00208-016-1462-6. URL `https://arxiv.org/abs/1404.2110`.

Weixu Su. Problems on thurston metric. American Institute of Mathematics workshop problem list, 2014. URL `https://aimath.org/pastworkshops/lipschitzteichproblems.pdf`. Problem 4.3 attributed to M. Kapovich; document dated April 21, 2014; problems originated at the 2012 AIM workshop; accessed 1 September 2026.

Weixu Su. Problems on the thurston metric. In Athanase Papadopoulos, editor, *Handbook of Teichmüller Theory, Volume V*, pages 55–72. European Mathematical Society, 2016. doi: 10.4171/160-1/3. URL `https://ems.press/content/book-chapter-files/23469`.

Dennis Sullivan. Related aspects of positivity in riemannian geometry. *Journal of Differential Geometry*, 25(3):327–351, 1987. URL `https://www.math.stonybrook.edu/~dennis/publications/PDF/DS-pub-0082.pdf`.